

Library Management System – Project Report

Introduction

The Library Management System is a simplified digital solution designed to manage a library's book inventory, track loan activity, and monitor overdue returns. This system is implemented using a relational database to ensure efficient data storage and retrieval.

Abstract

The project demonstrates a basic yet functional digital library environment using SQL and MySQL Workbench. It involves designing an optimized schema, populating tables with sample data, automating due dates using triggers, and generating meaningful insights using views and reports. This project lays the groundwork for scalable library systems.

Tools Used

- MySQL Workbench – for creating, managing, and testing the database
- SQL – for defining schema, manipulating data, and writing queries

Steps Involved in Building the Project

1. Defined schema with five tables: Authors, Books, Members, Loans, and BookAuthors (bridge for many-to-many).
2. Inserted 10 test records into each table to simulate actual data.
3. Handled many-to-many relationships between Books and Authors.
4. Created views:
 - borrowed_books – for active book loans
 - overdue_books – for late returns
5. Implemented a trigger to set the due date to 14 days from the loan date if not provided.
6. Executed SQL report queries for:
 - Number of books borrowed per member
 - Most borrowed books
 - All currently borrowed and overdue books

Conclusion

The project delivers a working prototype of a Library Management System with essential features such as book tracking, overdue monitoring, and usage analytics. Built entirely on SQL, it is extendable and serves as a strong foundation for more advanced systems.

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